DO NOW — RECALL

# Von Neumann and Harvard architectures

Name: _____ Class: _____ Date: _____

**1. What is the ideal completion cycle for four independent instructions in a three-stage pipeline?**

**2. What does MDR contain during an instruction fetch?**

**3. What happens in the Decode stage?**

**4. Calculate Sum and Carry-out for a full adder with A=1, B=0, Carry-in=1. Explain in binary.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Cycle 6 (=4+3−1).

### 2. Model answer

The instruction being transferred from memory.

### 3. Model answer

The CU identifies and interprets the opcode/operands.

### 4. Model answer

1+0+1=2 decimal = $10_2$, so Sum=0 and Carry-out=1.

### Vocabulary — know these terms

Von Neumann — Architecture with shared memory space for instructions and data.

Harvard — Architecture with separate program and data storage paths.

bottleneck — Limit caused by a shared resource or transfer path.

instruction cache — Fast cache storing instructions.

hybrid architecture — Combines features of different architectural approaches.

NEW LEARNING

# Read, annotate and explain

### Von Neumann organisation

Instructions and data share the same address space and a pathway; access can compete for bandwidth.

### Harvard organisation

Instruction and data memories/address spaces are separate, allowing independent access paths.

### Recognise contemporary designs

Many processors have aspects of both, such as a unified main memory and separate instruction/data caches (modified Harvard).

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### A CPU needs to fetch an instruction while reading its operand. Explain possible contention in a Von Neumann design.

### Predict first

The instruction fetch and data read may both need the same memory transfer pathway.

### Explain and check

The requests may have to be scheduled rather than served simultaneously; this is related to the Von Neumann bottleneck.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

State where instructions and data are stored in Von Neumann and Harvard architectures.

### Question 2

A CPU frequently fetches an instruction while reading data. Explain a potential Von Neumann limitation and Harvard advantage.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

A modern CPU uses unified main memory but separate L1 instruction and data caches. Explain why it cannot be described as a purely classical Harvard or purely classical Von Neumann design.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Every modern processor is exclusively a pure Von Neumann or pure Harvard machine.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.